Phu Nguyen

For this lab, I thought regression and classification were basically the same thing with different names. But after going through it, I see the difference is just what you're trying to predict. Fare is a number, and Survived is just yes or no. The part that helped me most was the metrics. MSE works for Fare because the prediction is almost never exactly right, so we check how far off it is. Accuracy works for Survived because it's either right or wrong. I also found it funny that logistic regression has "regression" in the name but it's actually for classification. The .coef_ part was interesting too, because it shows how much Fare changes when Age or Pclass goes up by 1. We also had to turn Sex into 0 and 1, since the model can't use words like male and female. And we just dropped the rows with missing Age, which was quick, but I guess we lose some data that way. Overall this lab helped me understand how to pick the right model and metric for the problem, and I can see why that's the first thing to figure out in ML.

Cyndy Ndong

During Lab 06 ITAI 1371, I focused on understanding and applying the core differences between regression and classification workflows using the Titanic dataset in Google Colab. My work began with data preprocessing, where I cleaned the dataset by dropping missing values from the Age feature and converting the Sex feature into numeric format by mapping male and female to binary values. Once the data was preprocessed, I separated the features and target variables, using train_test_split to create separate training and testing sets for both model types. This hands-on process highlighted how critical clean data types and proper feature representation are before passing data to machine learning algorithms.

For the regression portion, I implemented a Linear Regression model to predict continuous fare prices and evaluated its performance using Mean Squared Error, which helped me understand how continuous metrics penalize larger deviations. For the classification task, I imported and

built a Logistic Regression model to predict passenger survival, evaluating the model using accuracy score to measure correct categorical predictions. Throughout the lab, I encountered and systematically resolved several runtime errors, including incomplete code syntax, missing scikit-learn imports, unencoded categorical strings, and Name Errors caused by running notebook cells out of sequence. Successfully debugging these errors gave me a much deeper practical understanding of how data preparation and execution order impact model training in Python.